Date: 23/05/2026

Module Name: Version Control

<u>VERSION CONTROL ASSIGNEMNT</u>

Q1. What do you understand about Version Control?

Q 2. What is Terminal in software development

Q 3. Describe the Different types of version control systems

Q 4. Where version control is applied

Q 5. What are the benefits of version control?

Q 6. What do you understand about git?

Q 7. Provide a description of:

a. Features of git

b. Git basic concepts

c.Git architecture and git workflow

Q 8. what are the Basic commands to Configure GIT

Q 9. Which follow these stepsTo install Git on Windows.

Q 10. What do you understand about GitHub?

Q 11. Provide an explanation of:

a) The features of GitHub
b) Benefits of GitHub

c) GitHub account
d) Remote repository
e) Git repository commands as used in GitHub

Q 12.What are steps To follow to use GitHub effectively.

Q 13. Read the Following statement and answer by true if correct or false otherwise

Version control systems (VCS), including distributed versions like Git, are essential tools in software development that enable tracking changes over time, facilitate collaboration, and maintain a comprehensive version history. Through concepts like branching and merging, developers can work on separate features or fixes independently and later integrate them into the main codebase. Key commands like git pull to fetch and merge changes, git status to view repository state, and git branch to manage branches, all contribute to efficient project management by providing snapshots of the project at specific points and allowing seamless collaboration.

a. Version control systems (VCS) track changes to files over time, enabling collaboration and version history.

b. Distributed version control systems (DVCS) allow each user to have a complete copy of the repository with its full history.

c. Branching in version control allows developers to work on separate features or fixes without interfering with the main codebase.

d. Merging in version control combines changes from one branch or fork into another, integrating separate lines of development.

e. Commits in version control systems are snapshots of the entire project at a specific point in time, including all tracked files.

f. git pull fetches changes from a remote repository and merges them into the local repository.

g. git status displays information about the current state of the repository, such as modified files and branch status.

h. git branch is used to create, list, delete, or manipulate branches in a Git repository.

Theoretical assessment

Q 14. Match Git commands with its corresponding description

Command	**Description**
Git Commit	1. Manages connections to remote repositories.
Git Clone	2. Copies a repository from a remote source to your local machine.
Git Push	3. Fetches and merges changes from a remote repository to your local repository.
Git Pull	4.Removes files from the working directory and stages the removal for the next commit
Git Remote	4.Records changes to the repository
Git Status	5. Uploads local repository content to a remote repository.
Git Config	6. Sets configuration options for Git on your local machine.
Git rm	7.Shows the current state of the repository, including tracked/untracked files and changes

Q 15.Read the following sentences and Fill the gap with the missing word.

1. The command git __________ is used to create a new branch in Git.

2. A Git __________ is a pointer to a specific commit in the repository's history.

3. git checkout __________ is used to switch between branches in Git.

4. To list all branches in a Git repository, you can use git branch __________.

5. A Git __________ is a copy of a repository that lives on your computer instead of on a website's server.

6. git merge __________ is used to integrate changes from one branch into another.

7. git remote __________ is used to manage connections to remote repositories.

8. git push origin __________ is used to push the changes of the current branch to a remote repository.

9. A centralized version control system offers ………………………….a way to collaborate using a central server.

10. Distributed version control system allows ………………… management branching and merging.

Q16. Read this statement and answer by true if correct or false otherwise

Git Bash is a command-line interface for interacting with Git, a version control system. The git init command initializes a new, empty repository, while git clone is used to clone an existing remote repository. The git status command shows the state of the working directory, including untracked files, and git version displays the installed Git version.

a) Git bash is one among version control system that exist

b) Git init is used to clone remote repository

c) Git status is used to initialize an empty repository

d) In git you can view untracked files by using git version command

Q17. Mugabe XY holds the position of Senior Developer at Innovate Company Ltd, situated in Ruhango District. He has delegated a project to four developers, tasking them with designing a web application featuring various forms: login, student registration, course registration, and book registration, all using HTML. However, due to geographical constraints and other commitments, the developers found it challenging to collaborate effectively on the project. Consequently, the Senior Developer opted to assign tasks to each developer individually and remotely, recommending them to work autonomously on their designated tasks by referring to this repository structure.

├── login.html

├── student_registration.html

├── course_registration.html

├── book_registration.html

├── README.md

└── .gitignore

As one of the developers at the company, you have been assigned the following task:

Create a remote repository on GitHub using your full name as the repository name.

Clone the repository to your local computer.

Within the repository, create the required forms mentioned above for subsequent commits.

Q18. Describe the operations of these commands:

a) git status command
b) git add command
c) git reset command
d) Rm command

Q19. What do you understand about commit message?

Q20. What are the best practices for creating a commit message, operations related

to git log?

Q21. Can you explain the operations involved in the `git commit` command in Git?

Q23. What are the steps to commit file changes to a Git local repository?

Q24. 1. Match the Git branch operations with their corresponding commands?

Operation	Command
Create branch	a. git branch <branch-name>
List branch	b. git branch
Delete local branch	c. git branch -d <branch-name>
Delete remote branch	d. git push origin --delete <branch-name>
Switch branch	e. git checkout <branch-name>
Rename branch	f. git branch -m <old-branch-name> <new-branch-name>

Q25. Complete the sentence:

I. Before staging a file, you must check all.........files and.........files.

II. To create a new branch in Git, you use the command git __________ <branch-name>.

III. To list all branches in a Git repository, you use the command git __________.

IV. To delete a local branch, you use the command git __________ <branch-name>.

V. To delete a remote branch, you use the command git __________ origin --delete <branch-name>.

VI. To switch to a different branch, you use the command git __________ <branchname>.

VII. To rename a branch, you use the command git __________ <old-branch-name> <newbranch-name>.

Q26. How do I add files to a commit?

✓ $ git stage

✓ $ git commit

✓ $ git add

✓ $ git reset

Q27. How to save the current state of your code in git?

✓ By validating the modifications staged with $ git commit

✓ By adding all the changes and staging them with $ git stage

✓ By adding all the changes and organizing them with $ git add

✓ By creating a new commit with $ git init

Q28. Read the Following statement and answer by true if correct or false otherwise

The git add command is crucial for staging changes, whether they involve new files or modifications to existing ones, preparing them for the next commit. Once staged, the git commit command is used to permanently save these changes to the local repository, typically requiring a commit message for clarity. Additionally, the git commit command can be employed in the process of merging branches within Git, integrating changes from different lines of development.

a. The git add command is used to stage changes for the next commit.

b. The git add command can be used to stage both new files and modifications to existing files

c. The git commit command is used to permanently save changes to the local repository.

d. The git commit command requires a commit message to be provided.

e. The git commit command can be used to merge branches in Git.

Q29. As the project's owner, Sarah wants to tweak a few features and has assigned her coworkers

assignments to do so. She begins by creating a new Git repository and a branch called "feature_branch." By adding a new file with the name new_file.txt, Alex makes modifications to the project files. Emily verifies that the repository is functioning properly, adds new_file.txt to the staging area, and then commits the modifications with the message "Add new_file.txt."

David makes a switch to the main branch, merges the updates from "feature_branch," and clears up any conflicts. Finally, Sarah, the repository's owner, removes the "feature_branch" following a successful merge. They handle the project's branches, use the staging area properly, and commit file changes to the local Git repository.

Q30. What do you understand about the following term:

a. Fetch

b. Pull

Q31. n you explain the operations involved in the `git fetch` command in Git?

Q32. Can you explain the operations involved in the `git pull` command in Git?

Q33. What do you understand about the term push?

Q34. Can you provide a description of the tags used in the `git push` command?

Q35.Which of these Git client commands creates a copy of the repository and a working directory in the client's workspace. (Choose one.)

a) update

b) checkout

c) clone

d) import

e) None of the above

Q36. True or False? In Git, if you want to make your local repository reflect changes that have been made in a remote (tracked) repository, you should run the pull command.

a) True

b) False

Q37. Now, imagine that you have a local repository, but other team members have pushed changes into the remote repository. What Git operation would you use to download those changes into your working copy?

a) checkout

b) commit

c) export

d) pull

e) update

f) a, b, and c

Q38. fill-in-the-blank space related to fetching, pulling, and pushing files in Git

a) To fetch changes from a remote repository without merging them into your local branch, you use the command `git __________`.

b) To fetch changes from a remote repository and merge them into your local branch, you use the command `git __________`.

c) To upload your local repository changes to a remote repository, you use the command `git __________ origin <branch-name>`.

d) To see the status of your local repository, including which branch you are on and any changes that are staged for commit, you use the command `git __________`.

e) To add all changes in your working directory to the staging area, you use the command `git __________ `.

f) To commit your staged changes with a message, you use the command `git __________ -m "Your commit message"`.

g) To create a connection to a new remote repository, you use the command `git remote __________ <name> <URL>`.

h) To view the details of your remote repository connections, you use the command `git remote ___________`.

Q39. Match the operations with corresponding commands

Operation	Command
Fetch changes from remote repository	a. git fetch
Fetch and merge changes from remote	b. git pull
Push changes to remote repository	c. git push origin <branch-name>
See the status of the local repository	d. git status
Add all changes to the staging area	e. git add .
Commit staged changes with a message	f. git commit -m "Your commit message"
Create a connection to a remote repo	g. git remote add <name> <URL>

Q40. ABC Software Solutions is a leading software development company, Alex and Sarah are two experienced developers who need to collaborate on a project aimed at resolving a critical bug in a mission-critical software system for a major client. The bug has caused disruptions in the client's operations, and immediate attention is required to rectify the issue. Alex takes on the responsibility of simulating updates from the client and making changes to a separate branch to identify the root cause of the bug. Additionally, Alex merges the client's changes into their branch to test potential fixes and improve the overall stability of the system. Throughout the process, Alex actively contributes by making modifications, committing them, and pushing their branch to the remote repository, ensuring that the bug fixes are well-documented and can be easily shared with the client. Sarah, an integral team member, communicates her additional changes to Alex, addressing specific edge cases and proposing enhancements to optimize the software system's performance. After pushing her branch to the remote repository, Alex ensures synchronization by fetching Sarah's updated branch and merging it into their own, incorporating the valuable contributions made by Sarah. ABC Software Solutions demonstrates their commitment to providing high-quality software solutions and meeting their client's critical needs by leveraging efficient collaboration, version control, and code management practices to swiftly address and resolve complex problems in their client's software systems.